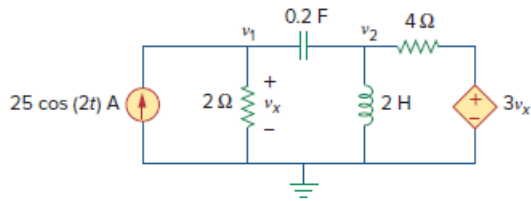


Problem

Using nodal analysis, find v_1 and v_2 in the circuit of Fig. 10.3.

Figure 10.3:



Step-by-step solution

Step 1 of 8

Determine the value of inductive reactance.

$$\begin{aligned} X_L &= j\omega L \\ &= j(2)(2) \\ &= j4 \Omega \end{aligned}$$

Determine the value of capacitive reactance.

$$\begin{aligned} X_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(2)(0.2)} \\ &= -j2.5 \Omega \end{aligned}$$

Step 2 of 8

Re-draw the circuit diagram with nodes.

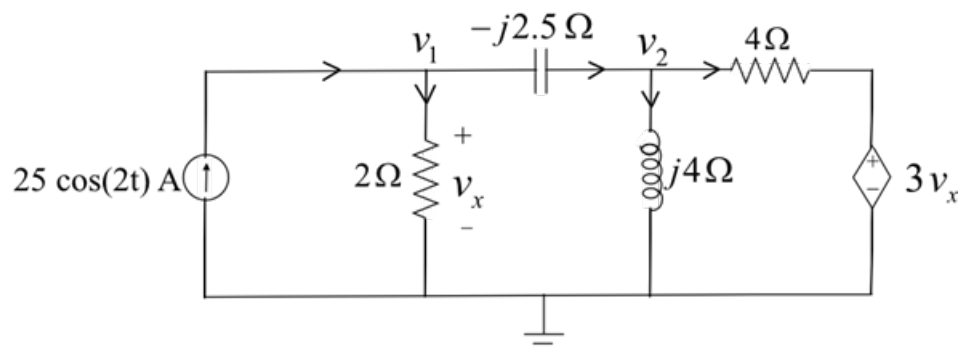


Figure 1

Step 3 of 8

Apply Kirchhoff's current law at node 1.

$$25 = \frac{v_1}{2} + \frac{v_1 - v_2}{-j2.5}$$

$$25 = 0.5v_1 + j0.4(v_1 - v_2)$$

$$v_1(0.5 + j0.4) - j0.4v_2 = 25 \dots\dots (1)$$

Apply Kirchhoff's current law at node 2.

$$\frac{v_1 - v_2}{-j2.5} = \frac{v_2}{j4} + \frac{v_2 - 3v_x}{4} \dots\dots (2)$$

Step 4 of 8

From figure 1, $v_x = v_1$

Substitute $v_x = v_1$ in equation (2).

$$\frac{v_1 - v_2}{-j2.5} = \frac{v_2}{j4} + \frac{v_2 - 3v_1}{4}$$

$$j0.4(v_1 - v_2) = -j0.25v_2 + 0.25(v_2 - 3v_1)$$

$$v_1(0.75 + j0.4) + v_2(-0.25 - j0.15) = 0 \dots\dots (3)$$

[Comments \(1\)](#)



Anonymous

incorrect, math is wrong. $(-4+7.5j)v_1 + (1.5-2.5j)v_2 = 0$

Step 5 of 8

Write equation (1) and equation (3) in matrix form.

$$\begin{pmatrix} 0.5 + j0.4 & -j0.4 \\ 0.75 + j0.4 & -0.25 - j0.15 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 25 \\ 0 \end{pmatrix}$$

Determine the value of Determinant of the matrix.

$$\begin{aligned} \Delta &= \begin{vmatrix} 0.5 + j0.4 & -j0.4 \\ 0.75 + j0.4 & -0.25 - j0.15 \end{vmatrix} \\ &= (0.5 + j0.4)(-0.25 - j0.15) - (-j0.4)(0.75 + j0.4) \\ &= -0.225 + j0.125 \end{aligned}$$

Step 6 of 8

Determine the value of Δ_1 .

$$\begin{aligned}\Delta_1 &= \begin{vmatrix} 25 & -j0.4 \\ 0 & -0.25 - j0.15 \end{vmatrix} \\ &= 25(-0.25 - j0.15) - (0)(0.75 + j0.4) \\ &= -6.25 - j3.75\end{aligned}$$

Determine the value of Δ_2 .

$$\begin{aligned}\Delta_2 &= \begin{vmatrix} 0.5 + j0.4 & 25 \\ 0.75 + j0.4 & 0 \end{vmatrix} \\ &= (0.5 + j0.4)(0) - (25)(0.75 + j0.4) \\ &= -18.75 - j10\end{aligned}$$

Step 7 of 8

Apply Cramer's rule, to determine the value of voltage at node 1.

$$\begin{aligned}v_1 &= \frac{\Delta_1}{\Delta} \\ &= \frac{-6.25 - j3.75}{-0.225 + j0.125} \\ &= 28.31 \angle 60.01^\circ \text{ V} \\ &= 28.31 \cos(2t + 60.01^\circ) \text{ V}\end{aligned}$$

Therefore, the value of voltage v_1 is $\boxed{28.31 \cos(2t + 60.01^\circ) \text{ V}}$.

Step 8 of 8

Determine the value of voltage, v_2 .

$$\begin{aligned}v_2 &= \frac{\Delta_2}{\Delta} \\ &= \frac{-18.75 - j10}{-0.225 + j0.125} \\ &= 82.56 \angle 57.12^\circ \text{ V} \\ &= 82.56 \cos(2t + 57.12^\circ) \text{ V}\end{aligned}$$

Therefore, the value of voltage v_2 is $\boxed{82.56 \cos(2t + 57.12^\circ) \text{ V}}$.